MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 12000011

(MODE SCALING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Mode Scaling Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assembly, Sight Current Generator, General specification for

DRAWINGS

U.S. Army, Frankford Arsenal

12000239	Sight Current Generator
10545455	Receptacle
12000011	Circuit Card Assembly, Mode Scaling

PACKAGING DATA SHEET

12000011	Packaging of Circuit Card Assembly, Mode Scaling

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Item definition.- This assembly is a single-ended printed circuit board consisting of a relay driver, a -7 Vdc power supply, a mode detection circuit, a range rate detection circuit and a triangle wave generator. The assembly is used to provide mode scaling and control circuit routing within the Sight Current Generator Drawing Number 12000239 in which it is assembled.

3.1.1 Major component list.- The components of this assembly shall be as specified on the Engineering Parts List (EPL) 12000011.

3.1.2 Government furnished property list. Not applicable.

3.2 Characteristics.

3.2.1 Performance.- The assembly shall be capable of performing the following requirements when assembled in accordance with Drawing Number 12000011.

a. Q1 and Q2 relay drivers shall control the K21 relay mounted in the sight current generator.

b. Operational amplifier AR1 shall control the -7 Vdc reference voltage for use in the sight current generator.

c. Transistor Q3 and operational amplifier AR4 shall detect whether the system is in radar mode or manual mode.

d. Operational amplifiers AR5 and AR6 shall detect whether the range rate signal is inbound or outbound.

e. The triangle wave generator (AR2 - AR3) shall generate an 18 volt P-P triangular wave within a period of approximately 1.00 milliseconds.

3.2.1.1 Input and output levels.

3.2.1.1.1 Input power.- Input power shall be supplied to the assembly as specified in Table I.

TABLE I - INPUT VOLTAGES

Pin	Voltage
24	+24 Vdc ± 2.0 percent
4	+15 Vdc ± 2.0 percent
14	-15 Vdc ± 2.0 percent
1 and 28	Ground

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3.2.1.1.2 Output levels.- The requirements specified in 3.2.1.1.2.1 through 3.2.1.1.2.6 shall be met when supplied with the specified voltage input.

3.2.1.1.2.1 Pin 3 output.- The output shall be as specified in a and b.

a. +13 Vdc ± 15 percent between pin 3 and pin one when -15 Vdc is applied between pin 19 and pin one.

b. -13 Vdc ± 15 percent between pin 3 and pin one when +15 Vdc is applied between pin 19 and pin one.

3.2.1.1.2.2 Pin 6 output.- The output shall be as specified in a and b.

a. Zero ± .01 Vdc between pin 6 and pin one when +15 Vdc is applied between pin 19 and pin one.

b. +15 Vdc ± 2 percent between pin 6 and pin one when -15 Vdc is applied between pin 19 and pin one.

3.2.1.1.2.3 Pin 11 output.- The output shall be a triangular wave with an amplitude from -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 1.00 ± .11 milliseconds.

3.2.1.1.2.4 Pin 12 output.- The output shall be as specified in a and b.

a. +13 Vdc ± 15 percent between pin 12 and pin one when zero Vdc is applied between pin 21 and pin one.

b. -13 Vdc ± 15 percent between pin 12 and pin one when +24.0 Vdc ± 15 percent is applied between pin 21 and pin one.

3.2.1.1.2.5 Pin 13 output.- R6 shall be varied between 6.8 and 7.2 Vdc then adjusted as necessary to obtain -7.0 Vdc ± .01 percent between pin 13 and pin one. The output shall not vary more than ± .1% between pin 13 and pin 1 during temperature extremes of -40°F to +160°F.

3.2.1.1.2.6 Pin 20 output.- The output shall be as specified in a through d.

a. +24 Vdc ± 15.0 percent between pin 20 and pin one with zero Vdc applied between pin 26 and pin one.

b. Zero ± .02 Vdc between pin 20 and pin one with +15 Vdc applied between pin 26 and pin one.

c. +24 Vdc ± 15 percent between pin 20 and pin one with +24.0 Vdc ± 5.0 percent applied between pin 23 and pin one.

d. +24 Vdc ± 15 percent between pin 20 and pin one with +24.0 Vdc ± 5.0 percent applied between pin 21 and pin one.

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3.2.1.2 Environmental.- The assembly shall meet the environmental requirements of specification MIL-C-50785.

3.2.2 Physical characteristics.- The physical characteristics shall be as specified in drawing 12000011 and the engineering parts list.

3.3 Design and construction.

3.3.1 Production drawings.- This assembly shall be fabricated in accordance with the requirements of specification MIL-C-50785, drawing 12000011 and associated lists.

3.3.2 Standards of manufacture.- The standards of manufacture shall be those specified in specification MIL-C-50785.

3.3.3 Workmanship.- The workmanship shall be as specified in specification MIL-C-50785.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.

4.1.1 Responsibility for inspection.- The responsibility for inspection shall be as specified in specification MIL-C-50785.

4.1.2 Special tests and examination.- The assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspection.- The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-50785.

4.2.1 Test equipment.

4.2.1.1 Power supply (PS1).- The power supply shall be +15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.2 Power supply (PS2).- The power supply shall be -15 ± 0.1 Vdc, 100 milliamperes (ma) or greater, ripple less than 10 millivolts (mv) rms or 100 mv peak-to-peak.

4.2.1.3 Power supply (PS3).- The power supply shall be +24 Vdc ± 2.0 percent, with regulation to ± .01 percent, 10 ma or greater.

4.2.1.4 Resistors.- Two resistors are required with a rating of 1100 ohms ± 5.0 percent, 2.0 watts.

4.2.1.5 Switches.- Four double pole, single throw are required.

4.2.1.6 Digital voltmeter (DVM).- The digital voltmeter shall have an accuracy and resolution of .01 percent or better from zero to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

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4.2.2 Functional test.

4.2.2.1 Test connections.- Power supplies PS1, PS2 and PS3 and resistors shall be connected to the assembly under test in accordance with Figure 1 using receptacle 10545455.

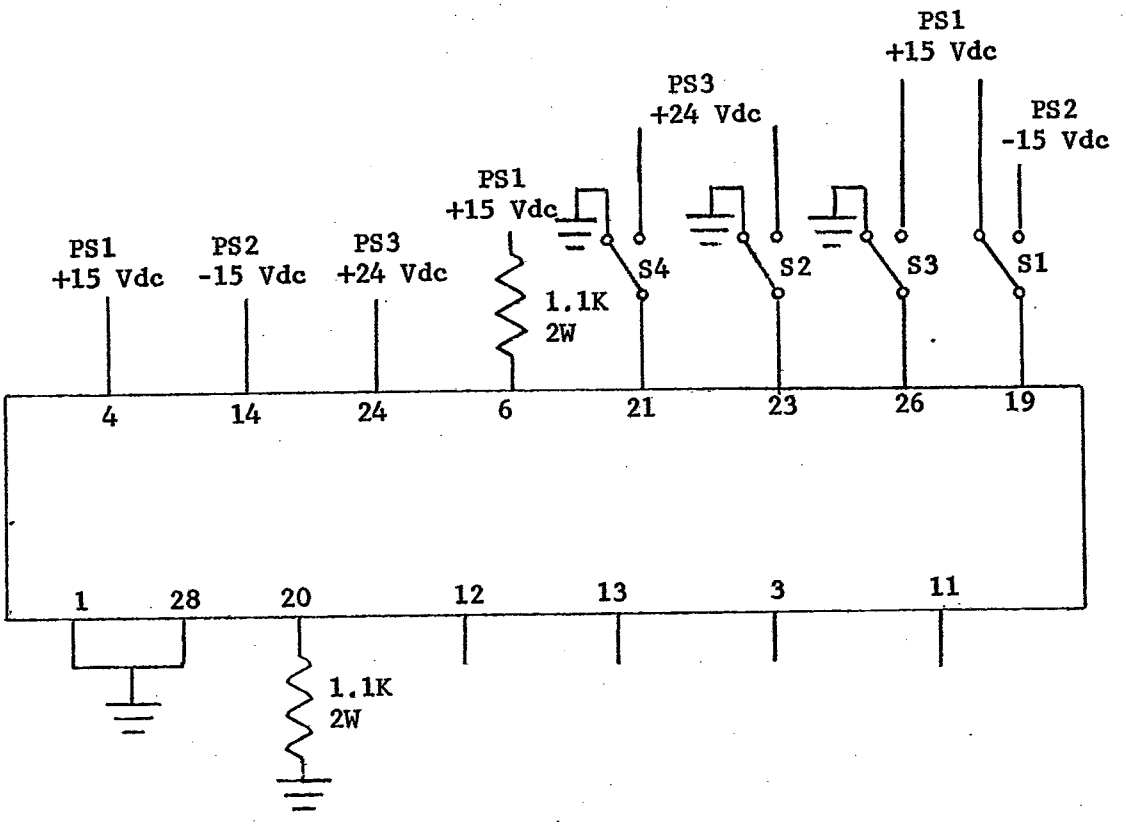


Figure 1. Test Connections

4.2.2.2 Test procedure.- Compliance with the requirements of 3.2.1.1.2 through 3.2.1.1.2.6 shall be verified as follows:

a. With switch S1 set to the -15 Vdc position, the voltage measured at pin 3 with the DVM shall be +13 Vdc ± 15 percent.

b. With switch S1 set to the +15 Vdc position, the voltage measured at pin 3 with the DVM shall be -13 Vdc ± 15 percent.

c. With switch S1 set to the -15 Vdc position, the voltage measured at pin 6 with the DVM shall be +15 Vdc ± 2 percent.

d. With switch S1 set to the +15 Vdc position, the voltage measured at pin 6 with the DVM shall be zero +0.1 Vdc.

e. With the oscilloscope connected to pin 11, the voltage shall be a triangular wave with an amplitude from -8.47 ± 1.03 Vdc to +8.47 ± 1.03 Vdc within a period of 1.00 ± .11 milliseconds.

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f. With switch S4 off, the voltage at pin 12 measured with the DVM shall be +13 Vdc ± 15 percent.

g. With switch S4 on, the voltage at pin 12 measured with the DVM shall be -13 Vdc ± 15 percent.

h. With the DVM connected to pin 13, the output shall be varied between 6.8 and 7.2 Vdc by adjusting R6 and then set at a voltage of -7 Vdc ± .01 percent. This output shall not vary more than ± .1% during temperature extremes after being set at -7 Vdc ± 0.01% at room ambient.

i. With switches S4, S2, and S3 set to the off position, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

j. With switch S3 set to on and S4 and S2 off, the voltage measured at pin 20 with the DVM shall be zero ± .02 Vdc.

k. With switch S2 and S3 set to on and S4 off, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

1. With switch S4 and S3 set to on and S2 off, the voltage measured at pin 20 with the DVM shall be +24 Vdc ± 15 percent.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC Form 1029, 12000011.

6. NOTES

6.1 The notes specified in the general specification MIL-C-50785 are applicable to this specification.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1285-A430

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